Imperial College London

Department of Aeronautics

Wing-In-Ground Effect 2017

Group Design Project

Individual Technical Report

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Abstract

This report summarises my role as one of the Project Coordinators of the Wing In Ground-effect Group Design Project. My tasks encompassed team leadership and management, as well as quantitative market, sea-state, aircraft profitability, and sales projection analysis. On the management side, we implemented an Agile-inspired management framework and formed specialist teams to tackle difficult design tasks. My analysis economically and practically validated our choice of East Asia as the target market, as well as providing a number of operating constraints. I also determined the mission capabilities and aircraft lifespan profit forecasts based on comprehensive vehicle cost and operator profit models. Ultimately my analysis showed that, whilst the implementation of a Wing In Ground-effect may be economically viable in East Asia, the large investment risk and existence of alternative regional transit systems in abundance mean that it is unlikely in the near future.

Contents

List of Tables

List of Figures

1 Introduction

This report is a summary of my activities as one of the Project Coordinators for the 2017 Wing In Ground-effect Group Design Project (WIG GDP). The task of the Project was to design and evaluate the efficacy of a modern WIG in a market of our choice. My roles included Group leadership and management as well as conducting quantitative analysis for the market, sea-state, aircraft profitability, and sales projections.

2 Leadership and Management

The WIG GDP task was highly technical and subject to a 5-week time constraint, with a prescribed, subject-based team structure. This kind of leadership presents an unique challenge; numerous historical examples of innovative technical initiatives, such as Edison's Invention Factory and AT&T's Bell Laboratories, show that a flexible working environment is conducive to success. In this section I outline how myself and fellow project coordinator Aymeric Pouyat attempted to facilitate such an environment whilst meeting the time constraints of the project.

2.1 Leadership style

Of all leadership theories available, I find a combination of two particular theories to be a simple, yet effective, means of developing a leadership style. These are Vroom-Yetton-Jago Decision Making Theory [1], and French and Raven's Bases of Power [2].

The premise of Vroom-Yetton-Jago Decision Making Theory is that the best leadership style is grounded in the situation. The 'styles' in this theory vary linearly from highly autonomous "autocratic" decision making to entirely passive "collaborative" decision making [1]. In the case of the WIG GDP, a greater sense of flexibility and inclusion was fostered using a more collaborative leadership style. This also allowed the group to make decisions independently, with myself and Aymeric acting only as mediators for difficult decisions.

French and Raven's eight Bases of Power categorise the means of persuasion available to a leader. The most relevant of these for our were referent and informational power. Referent power is generated via shared affiliations with members of the group [2]. For us, this was our shared affiliation to the College and Department. Specifically, the fact that the GDP was a University assessment was a useful means of encouraging swift, focused action. Informational power is generated via the possession of information by the leader that is desired by followers. As project coordinators, it was necessary for Aymeric and I to have an overview of the design at all times, which allowed us to motivate design decisions. However, in general, complex decisions were made via a collaborative process.

2.2 Overview of Management

The basis of our management style was the need for flexibility within a time-constrained working environment. To this end, we adopted the team networking software platform 'Slack' to augment the working environment. This, coupled with Google Drive, provided a central location for file storage, timetabling, and team announcements, and allowed team members to work remotely more readily. Initially I envisaged a mostly online working environment, however the level of communication in this ultra-flexible forum was unsatisfactory in the face of complex design interactions. As such, to assist in the clear provision of information, we conducted daily morning meetings in which we requested an overview of progress from each team. This management structure was partly inspired by the innovative Agile management philosophy. Developed in 2001, Agile is a philosophy for software development in the 21st century [3]. Among other things, Agile emphasises the use of self-organising, highly adaptive teams, and constant cross-collaboration between the technical and managerial aspects of a business [3]. In one popular Agile framework called 'Nexus', the working group conducts daily 'Scrums' to update the team on progress [4], which was the format of our morning meetings. We also called additional group meetings, when required, to discuss especially difficult or consequential design decisions.

Whilst mostly effective, this management framework was occasionally insufficient to tackle especially difficult design interactions - I discuss this in more detail later. The other major challenge that we faced was that, despite our best efforts, it was an ongoing struggle to keep everybody informed of important design decisions. This was mostly due to the failure of numerous group members to attend meetings and/or use Slack to communicate, despite being asked to do so on several occasions. Thankfully, the problems caused by this communication failure amounted to trivialities in the scope of the Project, all of which were promptly resolved following our weekly presentations to the Project supervisors.

In addition to managing the Group and dealing with design conflicts, Aymeric and I had a number of administrative duties. I have attached a list of my personal administrative duties at Appendix A.

2.3 Dealing with Design Interactions

Affects... / **Design parameter**	*Fuselage*	*Hull*	*Wing*	*Tail*	*Engines*	*Surface-effect*	*Systems*
Fuselage		Placement	Size	Size	Power	Efficacy	Placement
Hull	Stability		Placement	Size	Power	Power, efficacy	Placement
Wing	Structure			Size		Power	Placement
Tail	Structure						Placement
Engines	Structure			Placement		Power	Placement
Surface-effect				Size	Power		Placement
Systems					Power		

Figure 1: Design dependency matrix.

Figure 1 shows the design dependencies of the major elements of our WIG. As is evident, there is a huge need for a multi-disciplinary approach to WIG design. To this end, we created functional sub-groups that we called 'Guilds' to deal with particularly complex design interactions. Ultimately we formed four Guilds, which each had their own design purpose:

1. Lift Guild - to design the lifting surfaces of the WIG;
2. Take-Off Guild - to design the Power Augmented Ram (PAR) and hull systems to overcome take-off 'hump-drag;'
3. Mission Guild - to evaluate the economic performance of the WIG, and re-develop the mission profile for maximum profit (this process is detailed in the 'Mission optimisation' section of this report);
4. Tail Guild - to design the combined tail and engine systems.

Each of these Guilds was formed by combining members of each FATT into functional teams. This 'matrix structure' was inspired by the Agile management framework, which is described in [5]. However, this structure differed from our team structure in that, for us, the Guilds both performed functional roles in their own right, contrary to the original model. Forming our Guilds directed team focus and personnel towards critical design tasks and likely saved us considerable time overall.

2.4 Team performance metrics

In order to continuously improve group performance we measured performance via the quality of weekly presentations given to the supervisors. This was achieved using presentation assessment sheets that I analysed and published on a public leader-board each week. Most groups saw a general improvement in their scores over the course of the project and the feedback was useful in developing better technical solutions and improving presentation skills.

3 Baseline Configuration and Market Analysis

The first task in the Project was defining the target market and baseline configuration for the WIG. By its nature as a pseudo-marine craft, the market choice and configuration for a WIG are closely linked; larger aircraft can operate in more diverse conditions, but attract greater investment risk. The ideal market would provide both favourable operating and investment conditions. Aymeric and I requested that each FATT conduct a literature review and construct a short proposal as to which they thought would be the best configuration and market combination. Following the review of these proposals, the Group decided to pursue a medium-size passenger craft, with the remaining task to identify the most suitable market.

3.1 Economic characteristics of WIGs

A comprehensive literature review by the entire Group yielded a series of economic characteristics of WIGs. These, and the corresponding market conditions that would exploit these advantages are shown in Table 1.

Table 1: WIG economic characteristics and corresponding exploitative market characteristics.

WIG economic characteristics	**Corresponding market characteristic**
Percentage of possible operating conditions linked to sea-state	Relatively calm seas
Less reliance on complex airport infrastructure; ability to use existing dock infrastructure	Relatively few airports plus existing ferry infrastructure
Faster but more expensive to travel on than ferries	Existing ferry market and demand for high-speed transport
Slower but cheaper to travel on than commercial aircraft	(Excess) demand for air travel
Lower CO_2 emissions than both aircraft and ferries	Environmentally conscious market
Aircraft profitability and investment risk both scale with size	Significant capital investment and growth potential

3.2 Market selection

The requirement for relatively calm seas quickly narrowed down the market choice to a few areas globally that have large, protected sea areas. These are shown in Figure 2. In order to select a target market from amongst these, I conducted an economic analysis of each region shown in Figure 2 based on the GDP of the countries therein. A summary of this analysis by region is in Table 2, and a full list of all countries considered in each region is included at Appendix B.

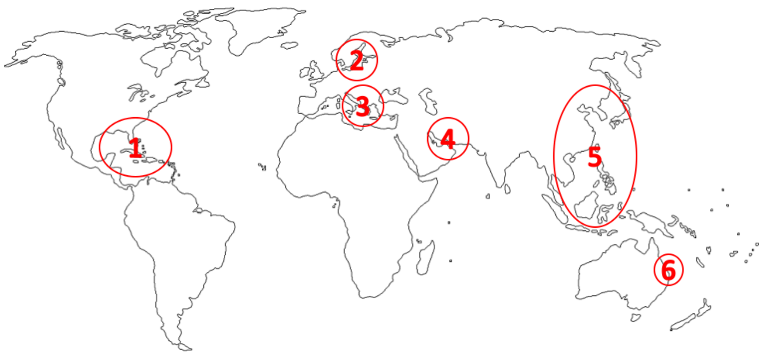

Figure 2: Potential markets with relatively calm seas.

Inspecting the figures in Table 2 it is evident that the two most attractive regions from an investment potential perspective are regions 4 (the Persian Gulf) and 5 (East Asia). Ultimately, Aymeric and I determined that, whilst the Persian Gulf has a greater average GDP per capita and average GDP growth rate, East Asia's significantly larger total and average GDP makes it a more attractive target market. Additionally, East Asia has the advantages of having an established ferry market, existing and popular high-speed transport systems such as bullet-train networks, and massive demand for short-haul air travel. Indeed, seven of the top ten most popular air routes in 2012 were East-Asian short haul routes [6] - a trend that appears unchanged in following years [7, 8]. A map and list of these routes is attached at Appendix C.

Table 2: Summary of regional economic analysis. All values are in 2016 USD, adjusted for Purchasing Power Parity, and sourced from [9]. Values in bold are the maximum for that category.

Region	**Total GDP** [$ USD $\times 10^{12}$]	**Average GDP** [$ USD $\times 10^{12}$]	**Average GDP per capita** [$ USD]	**Average GDP growth rate** [%]
1	31.73	2.27	17,700	1.28
2	9.95	1.11	36,100	1.79
3	4.32	0.72	22,600	2.43
4	5.33	0.67	**56,400**	**3.41**
5	**37.22**	**2.86**	34,400	3.25
6	1.19	1.19	48,800	2.90

4 Sea-State Analysis

Of significant concern in WIG design is the condition of the sea-surface from which the craft is operating, also known as the sea-state. Unless otherwise stated, most of the information and methods contained here are described in great detail by the World Meteorological Organization (WMO) in [10].

4.1 Introduction to sea-state meteorology

The most common measure of sea-state conditions is the significant wave height, H_S. Wave heights are always measured as the vertical distance from the crest to the trough of a wave (i.e. twice the wave amplitude). The significant wave height correlates well with wave height estimates made by shipboard observers, but is formally defined as the 'average height of the top one-third of the tallest waves' [10]. Since it is a statistical measure of wave height, a great amount of information about the sea-state can be inferred solely from H_S, such as the expected wavelength, wind speed, and other wave heights (e.g. the average or maximum wave height). In general, for a given sea region, the probability of encountering a particular H_S over the course of a year is distributed via a Weibull probability function. Further, for a given H_S (i.e. for a given sea-state), the distribution of actual wave heights follows a Rayleigh probability distribution. Thus, using hindcast wave height data for a given region, one may construct a statistical model for all of the sea-state parameters relevant to WIG design. An important point to note is that for all of my sea-state analysis I assumed the aircraft to be operating away from the coast where the effect of the sea-floor on wave parameters is negligible. The WMO defines this point as the depth at which $h > \frac{\lambda}{4}$, where h is the mean water depth and λ is the wavelength [10]. In deep water, the wave height is typically of the same order of magnitude as the wavelength, and wavelengths are rarely shorter than 10 times the wave height.

4.2 Data processing

Using regional wave hindcast data from [11] kindly supplied by Dr. Errikos Levis, I generated a series of Weibull Cumulative Distribution Functions (CDFs) for all of the regions of interest shown in Figure 2. The H_S CDFs for the regions in the target market are shown in Figure 3. In order to develop a more general dataset to be used for sea-state prediction, I then generated maximum, minimum, and median expected H_S CDFs for the East-Asian region, shown in Figure 4.

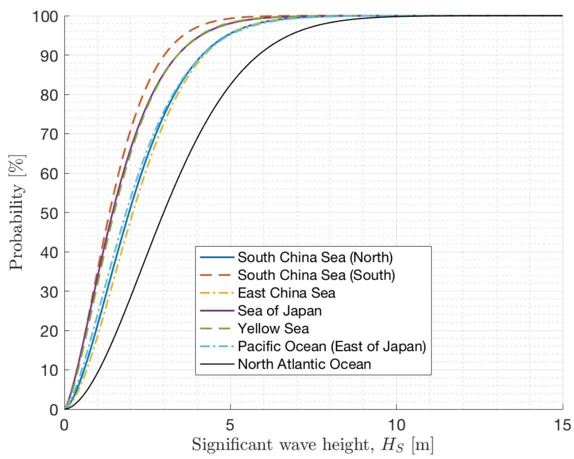

Figure 3: H_S CDFs for target market seas of interest. The North Atlantic CDF is included for reference.

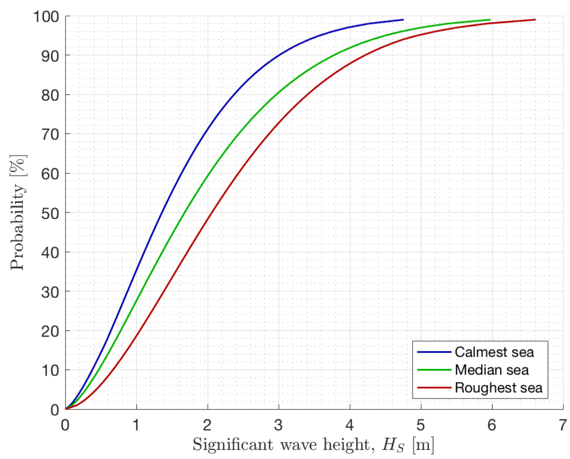

Figure 4: Statistically general H_S CDFs for the target market.

4.3 Applications in Analysis

4.3.1 Validation of Market Selection

Perhaps the most obvious application of sea-state analysis was to verify that the sea-states in the target market were sufficiently calm by comparison to known conditions. This is plainly shown by the horizontal position of the H_S CDFs in Figure 3. For example, the H_S CDF for the roughest sea in the target market - the East China Sea - is shifted significantly to the left of the North Atlantic CDF included for reference. Physically, this means that for any percentage of sea conditions encountered, the corresponding H_S for those conditions is smaller in the East China Sea than in the North Atlantic; i.e. the East China Sea is calmer. Indeed, the 95th percentile H_S in the East China Sea sea is 4.96m, whereas in the North Atlantic it is 6.77m.

Further, only two of all protected sea regions considered were calmer than the calmest East-Asian sea (the South China Sea); these were the Baltic Sea and the Persian Gulf. Thus our choice of East Asia as the target market was validated from a sea-state perspective.

4.3.2 Landing

Aymeric determined the worst case landing requirements for sea-planes from FAA regulations [12]; since the WIG is a special case of a sea-plane, we assumed these regulations to apply to our aircraft for this particular case. Specifically, the landing constraint was that $\lambda > 4l$ landing parallel to the sea-swell and $\lambda < \frac{l}{2}$ landing perpendicular to the sea-swell, where λ is the wavelength and l is the length of the WIG hull. According to Stokes' wave theory, a wave will break when $\lambda < 7H$, where H is the wave height, and since wavelengths are typically not shorter than $10H$ on the open ocean, the breaking condition was taken to be the worst-case relationship between wavelength and wave height [10]. For a conservative estimate of the wavelengths in given conditions, the height used to calculate the wavelength was the average wave height, which is slightly less than the significant wave height. Thus the procedure for determining the worst-case landing operability for our aircraft was, firstly, to apply the following equations in order:

$$\lambda = al; \qquad H_{avg} = \frac{\lambda}{7}; \qquad H_S = \frac{H_{avg}}{\sqrt{\frac{\ln 0.5}{-2}}} \tag{4.1}$$

Where a is the desired constraint relationship between the hull length l and wavelength λ (either $a = 0.5$ or $a = 4$ for landing parallel or perpendicular to the swell, respectively). The second part of the procedure was to determine the percentage of conditions for which H_S satisfies the constraints; for landing parallel to the swell this was the percentage of conditions with H_S lower than that computed, whereas for landing perpendicular to the swell it is the percentage of conditions with H_S greater than that computed. Ultimately, for a hull length of 23.079m, the perpendicular-to-swell landing constraint yielded a percentage of conditions less than 0.1% and so was insignificant. The parallel-to-swell landing constraint yielded 68.5%, 77.1%, and 86.4% of conditions as operable for the roughest, median, and calmest sea-profiles respectively.

4.3.3 Take-off

For take-off a more direct procedure was used. In order to have sufficiently low drag to allow acceleration at take-off only the hulls should be in the water. Thus, according to [13], for our configuration take-off is not possible in seas where $H_S > h_{hull}$, where h_{hull} is the total height of the hull (from the very bottom to the attachment point to the aircraft). This constraint yielded take-off as possible in 67.1%, 75.9%, and 86.4% of roughest, median, and calmest sea-profiles respectively. Since these values were slightly lower than those calculated for landing, they defined the overall operability constraints.

4.3.4 Cruise height

The requirement for the aircraft cruise height is that the bottom of the aircraft must not, at any point, make contact with the water. However, greater cruise heights reduce the efficiency of the aircraft as the lift and drag benefits of Ground Effect are reduced. For a given significant wave height, the maximum wave height that an observer will encounter in N total number of waves encountered may be calculated using the following formula from [10]:

$$H_{max} = 0.5 H_S \ln \sqrt{1.45N} \tag{4.2}$$

Further, N may be calculated via the following relationships:

$$c = \sqrt{\frac{9.81\lambda}{2\pi}}; \qquad N = \frac{1}{\lambda}(v + c)t \tag{4.3}$$

Where c is the wave group propagation speed, v is the aircraft cruise velocity, t is the flight time, and λ may be calculated from H_S by considering the wave breaking condition described in the Landing section above. Considering this relationship, we were able to reach a compromise cruise height of 3.1m from the mean sea-level. Applying a safety factor of 1.5, this correlated to a maximum wave-height of 4.2m. For the median sea profile, waves exceed this height in 51% of conditions.

4.3.5 Expected winds

Predicting the wave height from prevailing winds is a complex procedure requiring information about the wind speed, the area of sea over which the wind is acting (called the 'fetch'), and the duration for which the wind is acting. Thus, outside specific regions and weather conditions, it is virtually impossible to develop a generalised mathematical relationship between wind speeds and wave heights. However, it is common practice for meteorological agencies to use the empirical Beaufort Wind scale, which correlates wind speeds to average wave heights, to make wave height forecasts [14]. Thus, using this scale, we predicted that the wind speed in the worst case operating conditions (i.e. the 67.1% take-off constraint, for which the significant wave height is 2.74m) would be between 28 and 33 knots.

5 Profitability Modelling

My primary area of focus for the second half of the Project was developing the business case for the WIG by creating operator and manufacturer profit models.

5.1 Vehicle cost model

There are several established methods for predicting aircraft cost, usually based on the aircraft empty weight (ACEW), cruise speed, and other basic operating parameters. These include, for example, the DAPCA method [15], Torenbeek's method [16], and the Nicolai method [17]. All of these methods are based on some kind of curve-fitting of historical aircraft cost data to input parameters and thus all fall prey to the major hindrance in WIG design: a lack of available historical data. Indeed, even after contacting several manufacturers directly, I could only find cost data for one WIG (the Wingship 500) [18] and that too was a pre-production cost estimate. This lack of data made it impossible to judge the accuracy of any given cost estimation method. Without conducting a numerical analysis, it was clear from the literature review that of the three methods listed here the DAPCA method yielded the highest costs [19], and as such I judged it most appropriate to focus on refining this fairly conservative model for the purposes of WIG design. In essence, the DAPCA model, currently in its' 4th version, computes a series of required production hours as well as development, flight test, and materials costs based on the ACEW, cruise speed, and quantity produced. The first, critical step of updating this method was to update the constant multipliers, quoted for 2012 USD in [20], to 2017 USD. I did this by comparing the consumer price index for both years using a US Government tool at [21]. Towards the end of the Project, when we had a much clearer idea of the materials, manufacturing methods, and installed systems, I updated the DAPCA IV model to reflect the actual cost values anticipated for all of these elements. Thus the final vehicle cost model used the DAPCA method to compute engineering, tooling, quality control, development support and flight test costs, with the manufacturing, materials, engine, and systems costs based on actual data from the Structures and Systems/Propulsions teams.

5.2 Operating profit model

I developed an operating profit model from first principles by considering how a company flying WIGs commercially in the target market would operate. Table 3 shows all of the constant elements of the operating profit model and how they were determined. The model took as inputs the number of legs flown, the distance per leg, the estimated vehicle cost (from the modified DAPCA scheme), and the fuel used per mission. The parameters not estimated via DAPCA were calculated using a performance model of the aircraft that I developed in collaboration with the Systems/Propulsion and Flight Mechanics teams - this is described in more detail later. From these inputs, the operating profit model then calculated the number of missions possible per day without running over a 12h limit (flying at night was deemed to be too dangerous), the number of trips per mission, the number of tickets sold per trip (and hence revenue), and all of the crew and maintenance costs based on flight hours per day. It then multiplied these daily costs with the number of operable missions per year for an annual profit estimate.

Table 3: Constant elements of operating profit model.

Model element	Value	Source
Number of pilots	2	Convention
Number of attendants	3	FAA FAR regulations [22]
Pilot pay rate	USD $45 / operating hour	Comparison of Cathay Pacific pilot pay rates and ferry crew pay rates [23, 24, 25]
Attendant pay rate	USD $15 / operating hour	(As above)
Annual fraction of operable weather conditions	0.7595	Landing constraint, see above section
Daily operating hours	12	Assumption; average over one year
In-port turnaround time	10 minutes	Assumption
Refuelling turnaround time	1 hour	Assumption
Crew operating hours	1.15 $\times$ total flight hours	Cathay Pacific pilot pay rates [24]
Price of Jet-A1	1.352	[26] on 13 Jun 17
Passenger Load Factor	0.68	Linear extrapolation based on [18]
Ticket price	USD $0.23 per passenger per nautical mile	Japanese market study, see below
Maintenance cost; aircraft	USD $1125 per flight hour	[27]; IATA report
Maintenance cost; marine craft	10% of vehicle purchase cost per annum	[28]; Forbes report

5.2.1 Ticket prices: Japanese market study

In order to determine a reasonable WIG ticket price I conducted a detailed competitor analysis for the Japanese market. Japan has arguably the most developed transit infrastructure in the world; it boasts an advanced bullet-train network (the Shinkansen), a well established ferry system, as well as a great number of large domestic airports. Whilst there is evidently huge demand for transit to be exploited, the market is so saturated that any newcomers must have some kind of significant competitive advantage in order to survive. One of the key advantages that a WIG may offer is based on an intermediate ticket price between ferries and aircraft. In

Japan, this is complicated by the Shinkansen bullet train network, which represents additional competition. Thus the object of my competitor analysis was to develop a ticket price that would give a WIG operator an advantage in a super-competitive market. The analysis considered every long-distance ferry route in Japan from [29], all of the budget domestic flights from Tokyo listed on Google Flights and their flight distances calculated via [30], and every possible route combination from Tokyo on the Tōkaidō and Sanyō Shinkansen lines as listed at [31]. A summary is shown in Table 4. From the summary I deduced that a ticket price between \$0.20 and \$0.23 per nautical mile would yield a sufficient ticket pricing advantage over both conventional flight options and the Shinkansen for an operator to remain competitive.

Table 4: Summary of Japanese market competitor analysis. 'nmi.' stands for nautical miles.

Transport Mode	**No. routes considered**	**Average distance [nmi.]**	**Average ticket price [$ USD]**	**Average price per nmi. [$ USD / nmi.]**
Ferries	49	602	68.51	0.16
Budget domestic flights	33	393	86.51	0.27
Shinkansen	48	301	124.77	0.56

6 Economic Optimisation

6.1 Second iteration mission optimisation

Closely linked to profitability modeling, one of my goals was to refine our mission profile selection such that, for a future second iteration design, the WIG would be cheaper to produce and/or more profitable for the operator. The original intended method for this endeavour was to create a series of MATLAB functions that, on a crude level, would collectively design an entire WIG and evaluate its profitability. This program could then be optimised for maximum profits and/or minimum costs via a MATLAB optimisation routine. A flowchart of how this design program would work is attached at Appendix D. Unfortunately, this endeavour proved to be too ambitious; many of the design interactions were too complex to be modeled even on a crude level within a series of MATLAB functions and the ones that weren't too complex were extremely computationally expensive and time consuming. For example, a single run of the hull-design function that I developed with Wei It Kuan from the Aero/Hydro team took over 11 seconds, which after lengthy improvements was reduced to around 7 seconds. In the face of this adversity I opted to greatly simplify the design routine to be optimised; a flowchart of this simplified routine is shown in Figure 5.

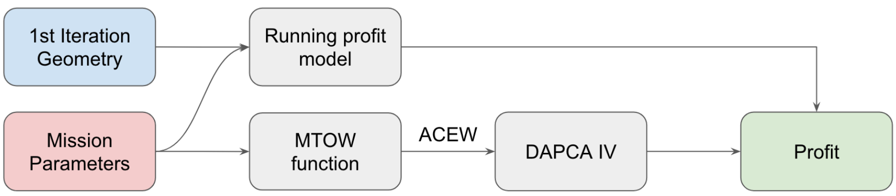

Figure 5: Flowchart of simplified WIG design routine for which to optimise.

For this simplified routine, the Passenger Load Factor and Number of Attendants in the running profit model were both modified such that they were a function of the passenger capacity of the aircraft. The Passenger Load Factor was assumed to be a linearly decreasing function of passenger capacity based on [27] and [18], and the number of attendants was defined via FAR regulations from [22]. The maximum take-off weight (MTOW) function used the methods used to calculate the MTOW and ACEW originally; see [32, 33] for more detailed discussions of the assumptions therein and [34] for how these were later refined. Because of its simplified nature, any output from this new design routine could only be considered valid if the inputs were in a restricted region about the first iteration design parameters. Subject to a series of constraints (Detailed in Appendix E), using the MATLAB 'fmincon' function the simplified design routine returned a converged optimisation. The output operator profit forecasts are plotted in Figure 6 and the corresponding mission profile parameters included Appendix F. Unfortunately, following several design delays, the Group made a collective decision to not pursue a second iteration design, which meant that this analysis went unused.

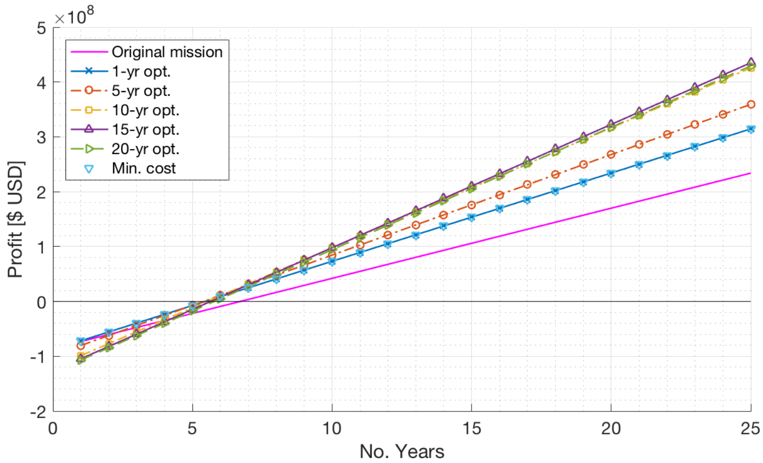

Figure 6: Operator profit forecasts for optimised mission profiles. Note that the minimum cost and 1-year optimal profit solutions are converged, as are the maximum profit solutions for greater timespans.

6.2 First iteration economic performance evaluation

In order to evaluate the sellability of our finalised first iteration design I applied the economic modeling techniques that I had developed for the second iteration mission selection. The objectives of this new analysis were to determine the most profitable mission profile that the aircraft was capable of, to determine a reasonable sales profit margin for the manufacturer of the aircraft, and, considering these, to determine the number of aircraft to produce to ensure a valid operator purchase value proposition.

6.2.1 Mission capabilities

In order to determine the mission capabilities of the aircraft I worked closely with the Flight Mechanics team to develop a more comprehensive aircraft performance model (see [34] for more details). This model took as inputs the required mission leg distance and fuel tank size and computed the maximum number of legs possible subject to the fuel constraint and the daylight flying hours constraint. Having completed this model, I then worked with the Systems/Propulsion team to match the actual aircraft fuel tank size to the constraints imposed by the daylight flying hours limit as well as the geometry of the aircraft. This was done because installing a fuel tank that would enable mission times greater than 12 hours would be to over-design the aircraft; a vehicle with such a large fuel tank would have to carry additional empty weight and, thus, the fuel consumption in 12 hours would be larger than an aircraft with a fuel tank designed to allow exactly 12 hours of flight. The fuel tank capacity was thus determined to be 28,000kg. The output of this mission capability model is shown in 7. In Figure 7 the capability front is the lower bound of both time and fuel constraints. Note that the design point is the 4 × 250 nmi. legs initially designed for, the practical time limit is a leg travel time of 2.5h, and the Power Augmented Ram (PAR) recharge time is 1.4h. From this plot it is evident that the final design is easily capable of outperforming the intended design point - indeed, must outperform in order to recharge the PAR system batteries! Additionally, for each mission profile I conducted a 15-year operator profitability analysis (assuming this to be a reasonable estimate of airframe lifespan, based on [18]); this is shown in Figure 8.

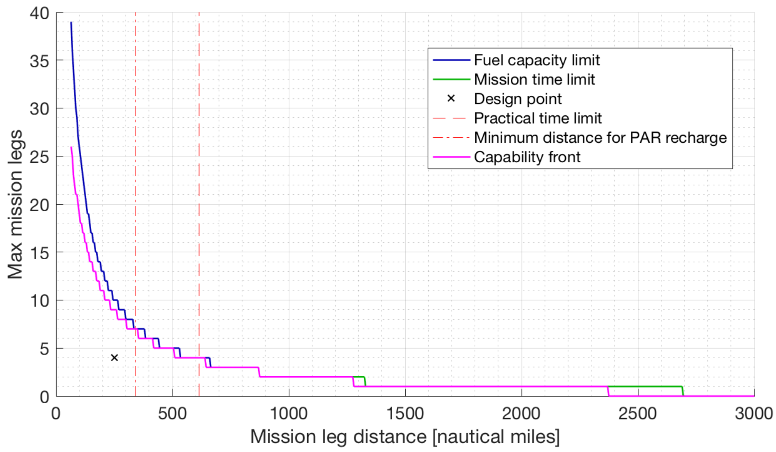

Figure 7: Capability front for the first-iteration subject to daylight flying hours and fuel tank size constraints.

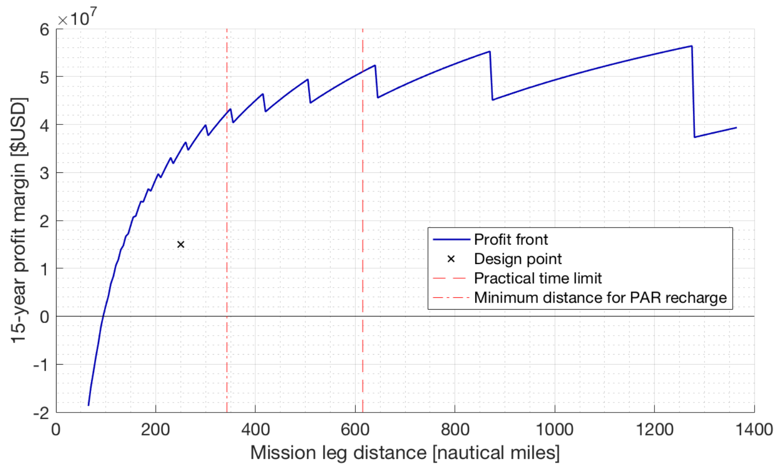

Figure 8: 15-year profit capability front of first-iteration design.

6.2.2 Sales profit margin analysis

In order to determine an appropriate vehicle sales margin I reviewed Airbus sales data [35, 36, 37, 38]. The objective was to compare the actual profits reported by Airbus Commercial Aircraft to that projected by multiplying actual aircraft deliveries each year by the list price of each aircraft. This analysis yielded profit margins of 5% and 95% for the actual profits and projected profits respectively. The large range of profit margins predicted may be due to Airbus bearing the cost of delayed deliveries and/or making losses on bulk deals on its larger and newer aircraft such as the A380. Considering that the 5% margin represented an actual profit figure reported by Airbus, I took this as being the value likely closest to the average aircraft sale profit margin. Given that early estimates showed the WIG as costing around 40%

of the list price of an A320 [36], I eventually settled on 10% as a reasonably sales profit margin for our WIG.

6.2.3 Number of aircraft to produce

Considering an operator flying the most profitable mission profile shown in Figure 8 and a vehicle sales profit margin of 10%, I was able to synthesise my performance, vehicle cost, and operator profit models to determine the number of aircraft to produce and corresponding vehicle sales profit margins. This is summarised in Table 5, and operator profit forecasts are plotted in Figure 9.

Table 5: Summary of predicted aircraft production and sales data.

15-year operator profit margin	No. vehicles to produce	Vehicle sale price [$ USD $\times 10^6$]	Manufacturer profit [$ USD $\times 10^6$]
0% (break-even)	72	54.7	358
50%	113	45.7	470
100%	179	39.2	638
200%	532	27.7	1474

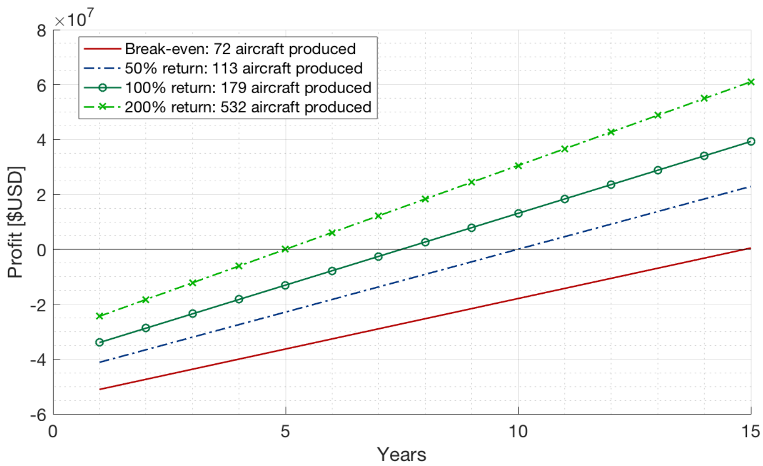

Figure 9: Operator profit forecasts for various numbers of aircraft produced.

7 Conclusion

Perhaps the most significant lesson I learned from my management role was that it is exceptionally difficult to lead a technical team in a pressured environment without possessing any source of legitimate power. This was compounded by the need for me to conduct my own analysis and thus I think that the main improvement to the management aspect of the project would be to implement a dedicated business team or at least to expand the size of the Project Coordination team. From a technical standpoint, I believe my analysis shows that there is a market potential for WIG sales and operations. However, the magnitude of the initial capital investment required to ensure operator profits represents a significant risk. Thus, the success of WIG implementation would require the synthesis of significant existing economies of scale from both a manufacturing and operating perspective. Given its size, this would be possible in the East-Asian market. Ultimately, however, I believe that the economic inconvenience of WIG research and development, coupled with the convenience of currently available regional transport systems means that, without great economic pressures (such as a monumental surge in jet fuel price or a financially crippling carbon tax), we are unlikely to see a successful commercial WIG design in the near future.

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Appendices

Appendix A List of personal administrative duties

- Emailing the supervisors weekly regarding the Friday briefing times;
- Creation of the weekly presentation master Google Slides file;
- Assignment and management of presentation order and timings;
- Creation of the presentation assessment sheets;
- Presentation assessment data entry and processing each week;
- Management of Slack channels and communications;
- Google Drive file management;
- Creation of the LaTeX Group Report template;
- Communication with relevant supervisors regarding management issues;
- Construction of Group Executive Summary; and
- Management of final Group Presentation creation;

Appendix B List of countries in each market region

Region	**Countries**
1	Belize, Colombia, Cuba, Dominican Republic, Haiti, Honduras, Jamaica, Mexico, Nicaragua, Panama, Puerto Rico, The Bahamas, USA, Venezuela
2	Denmark, Estonia, Finland, Germany, Latvia, Lithuania, Poland, Russia, Sweden
3	Albania, Croatia, Greece, Italy, Montenegro, Turkey
4	Bahrain, Iran, Iraq, Kuwait, Oman, Qatar, Saudi Arabia
5	Brunei, Cambodia, China, Hong Kong, Indonesia, Japan, Malaysia, Phillipines, Russia, Singapore, South Korea, Taiwan, Vietnam
6	Australia

Appendix C Popular East-Asian air routes

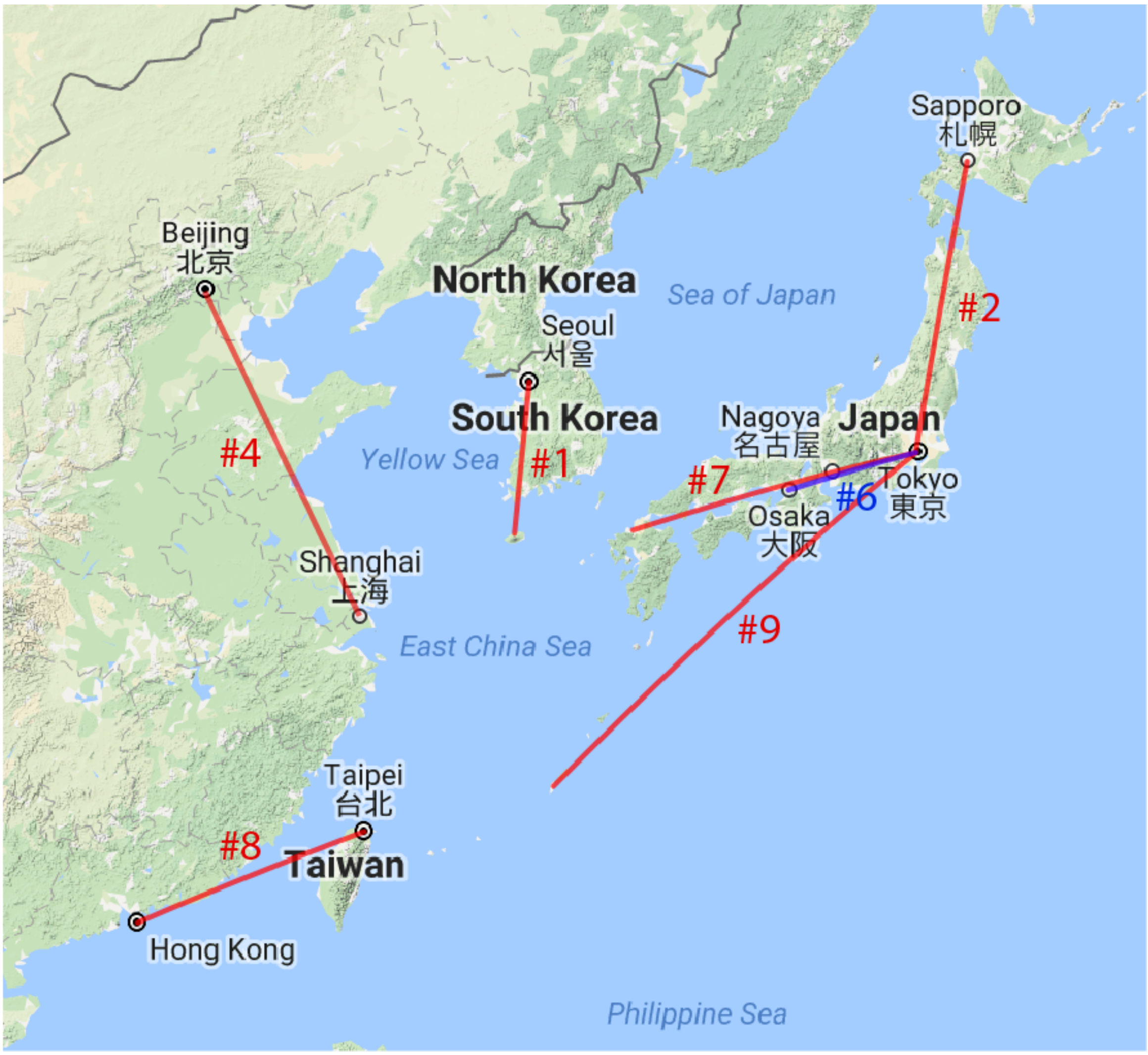
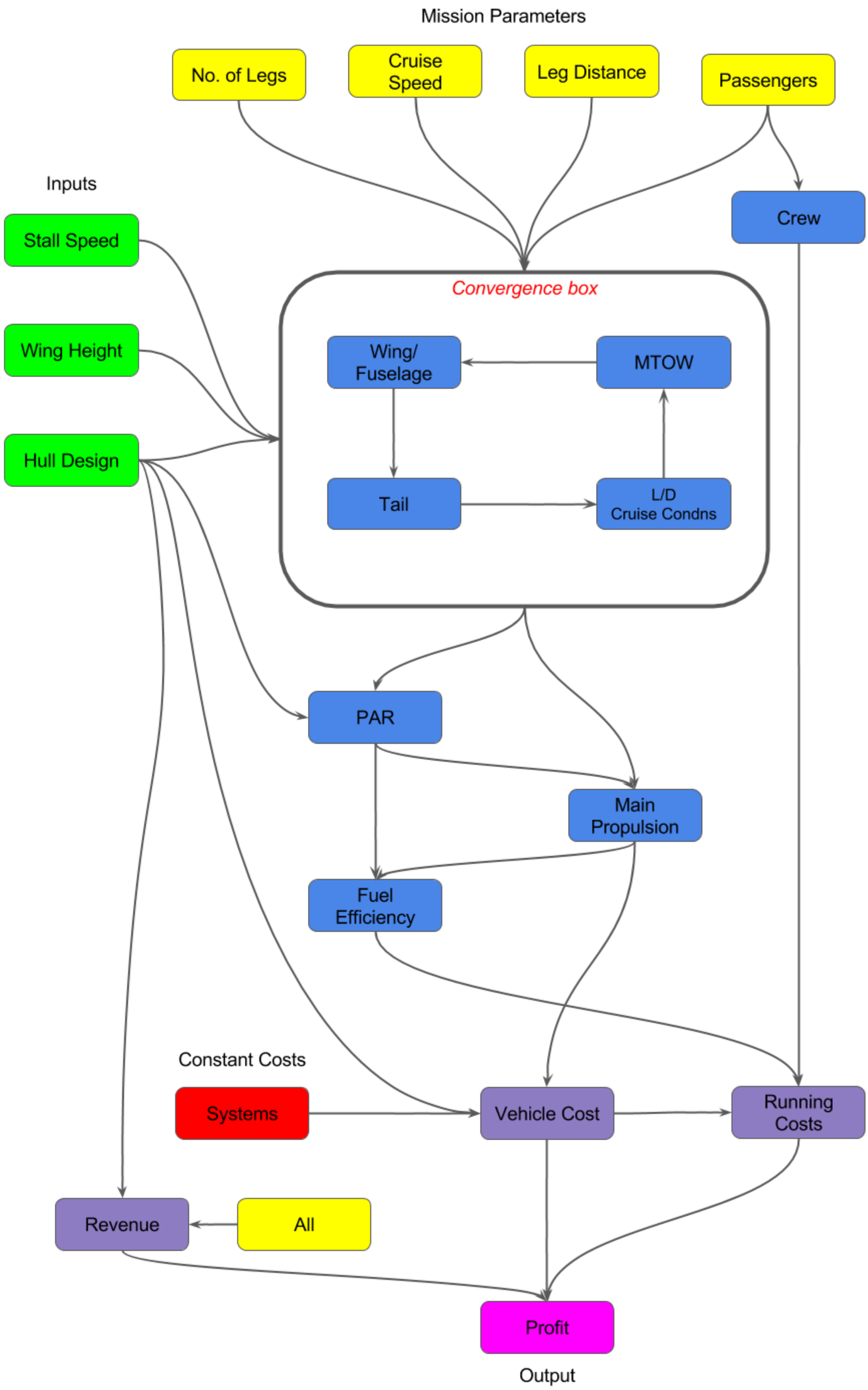


Figure 10: Map of popular East-Asian air routes and rankings.

Table 6: List of popular East-Asian air routes, world rankings, and passenger volume, from [6].

World Ranking (2012)	**Route**	**Annual Passenger Volume (2012) [millions]**
1	Seoul - Jeju	10.156
2	Tokyo - Sapporo	8.211
4	Beijing - Shanghai	7.246
6	Tokyo - Osaka	6.744
7	Tokyo - Fukuoka	6.640
8	Hong-Kong - Taipei	5.513
9	Tokyo - Okinawa	4.584

Appendix D Flowchart of intended 2nd-iteration optimisation design routine



Appendix E Table of second iteration mission optimisation constraints

Variable	Lower Bound	Upper Bound
Passenger capacity	75	225
Cruise velocity [kts]	164	364
Leg distance [nmi.]	100	800
Leg time* [mins]	15	150
Mission time* [hours]	0	12

* These were computed as functions of cruise velocity, leg distance, and, for mission time, the number of legs. As such they appeared as non-linear constraints in the optimisation.

Appendix F Table of output optimised mission profiles

	Mission Profile			
Optimisation Goal	**Passenger Capacity**	**Cruise velocity** [kts]	**Leg distance** [nmi.]	**No. legs**
1-year max. profit	165	300	350	3
5-year max. profit	195	302	348	3
10-year max. profit	225	344	302	4
15-year max. profit	225	364	279	4
20-year max. profit	225	364	268	4
Min. vehicle cost	165	300	350	3